

# GenAI Badge Notes

*Personal study notes — Google Cloud / Vertex AI*

## Generative AI

A type of AI technology that can produce various kinds of data — text, imagery, audio, and synthetic data. The key idea is that it creates new content rather than just classifying existing content.

## Artificial Intelligence

A branch of computer science focused on building systems that can reason and learn, with the goal of making machines that think and act like humans.

## Machine Learning

A subfield of AI that trains a model from input data. The trained model uses patterns from past data to make future predictions.

- **Supervised** — uses labelled data; the model learns from past examples to predict future values.
- **Unsupervised** — uses unlabelled data; the goal is discovery, seeing if data naturally falls into groups.
- **Deep Learning** — uses artificial neural networks inspired by the human brain. Can use both labelled and unlabelled data. Labelled data helps learn basic concepts; unlabelled helps generalise to new examples.

*GenAI is a subset of deep learning and can process unlabelled data using both supervised and unsupervised techniques.*

## Generative vs. Discriminative Models

- **Discriminative** — classifies or labels data points. Trained on labelled data. Output: a number, discrete class, or probability.
- **Generative** — generates new data instances. Output: natural language, image, audio.

Mathematically:  $Y = f(X)$  — Y is the model output, f is the function the model embodies, X is the input data (value files, image files, audio, etc.).

## How Robust is GenAI?

Trained on both labelled and unlabelled data across all data types to create a foundational model. In traditional programming you'd hardcode rules (e.g. to distinguish a cat from a dog). Neural networks just need a picture. GenAI goes further — it can generate its own content. Ask it what a cat is and it gives you everything it has learned.

## Transformers

The power of GenAI comes largely from transformer architectures. At a high level, a transformer consists of an **encoder** and a **decoder**.

## Hallucinations

Words or phrases generated by a model that are nonsensical or factually incorrect. Caused when the model hasn't been given enough data, making it more likely to produce unreliable outputs.

## Prompts

A short piece of text given to a large language model as input. Used to control the output of the model in a variety of ways. GenAI depends heavily on the quality of the input data and the prompt.

## Text Input/Output Modalities

- **Text → Text** — maps text to text. Example: language translation.
- **Text → Image** — trained on large datasets of images each captioned with a short text description.
- **Text → Video** — output is a video that corresponds to the input.
- **Text → 3D** — used in games or other 3D environments.
- **Text → Task** — wide range of actions, e.g. navigating a web UI or making changes to a document.

## Foundational Models

A large AI model pre-trained on a vast quantity of data, designed to be adapted or fine-tuned for a wide range of downstream tasks like sentiment analysis, image captioning, and object recognition.

*Vertex AI offers a Model Garden which includes foundational models.*

## GenAI for Code Generation

GenAI can help debug files of code, translate code from one language to another, and explain code line by line.